

A PROJECT REPORT ON:

ANALOG LINE-FOLLOWER ROBOT WITH DIFFERENTIAL STEERING

SUBMITTED TO:

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1.0 ABSTRACT

This report details the design and implementation of an autonomous Line-Follower Robot (LFR) utilizing a purely analog control system. The primary objective is to engineer a vehicle capable of navigating a black track on a white surface using Infrared (IR) sensors, an LM358 operational amplifier for signal processing, and an L293D motor driver. The system is designed to meet the specific requirements of the EEG 331 design competition, which mandates autonomous tracking across line widths of 20mm, 15mm, and 10mm.

The design focuses on hardware-level calibration through the tuning of potentiometers to set optimal threshold voltages for different lighting conditions and line thicknesses. The system's performance is evaluated based on tracking accuracy, completion speed, and the ability to recover from path deviations. This project highlights the effectiveness of analog feedback loops in achieving real-time responsiveness and deterministic behavior in autonomous robotics.

1.0 INTRODUCTION

Autonomous navigation systems represent a cornerstone of modern industrial automation. This project details the design and implementation of an Automated Guided Vehicle (AGV)—a robot capable of navigating a predefined path using a purely analog control architecture. In the context of modern logistics, such as automated warehousing and manufacturing assembly lines, AGVs facilitate the seamless movement of goods with high efficiency and minimal human intervention. Mastering the principles of analog tracking systems is essential for engineers, as these systems provide real-time responsiveness and environmental robustness without the overhead of complex microcontrollers.

The primary objective of this project is the development of a robot that utilizes analog signal processing to achieve autonomous line-following. Unlike digital systems that rely on pre-programmed software, this robot executes decision-making through the physical interaction of discrete hardware components. Specifically, the system employs LM358 operational amplifiers as voltage comparators and the L293D integrated circuit for motor power management.

2.0 OBJECTIVES

The main goals of this design project are as follows:

- **Design and Construction:** To design, build, and test a robot capable of autonomously following a predefined line using infrared (IR) sensors and analog circuits.
- **Analog Integration:** To successfully integrate sensing, decision-making, and actuation into a functional system without the use of a microcontroller.
- **Performance Reliability:** To ensure the robot can follow lines of varying thicknesses during the competition rounds: 20mm standard, 15mm challenge, and 10mm precision.
- **Optimization:** To maximize performance based on Tracking Accuracy (50%), Completion Speed (30%), and Recovery (20%).

3.0 SYSTEM DESIGN

The robot is designed using an analog closed-loop control system based on a differential drive platform.

3.3.1 System Components

- **Sensing (Input):** Two IR Sensor pairs (IR LED and photodiode) mounted at the front of the chassis.
- **Processing (Control):** An LM358 Dual Op-Amp acting as a comparator against a threshold set by 10k Ω Potentiometers.
- **Actuation (Output):** An L293D Motor Driver providing high-current power to DC Geared Motors.

3.3.2 Operational Mechanism

The robot maintains its course through differential steering. On a white surface, IR reflectance is high, keeping motors active. On a black line, light is absorbed, triggering the comparator to cut power to the corresponding motor, causing the robot to pivot back to center.

4.0 CALCULATIONS AND ENGINEERING DESIGN

4.4.1 Voltage Divider and Signal Acquisition

The voltage at the non-inverting terminal is given by:

$$V_{in} = V_{cc} \times \left(\frac{R_s}{R_s + R_{fixed}} \right)$$

For white surfaces, $R_s \approx 10k\Omega \implies V_{in} \approx 0.81V$. For black lines, $R_s \approx 500k\Omega \implies V_{in} \approx 7.5V$.

4.4.2 Threshold Calibration (V_{ref})

Reference voltage is set at the midpoint to provide maximum noise immunity:

$$V_{ref} = \frac{V_{max} + V_{min}}{2} \approx 4.15V$$

4.4.3 Kinematics and Velocity Analysis

Using 60 RPM motors and wheels with diameter $D = 65mm$:

$$v = \omega \times \frac{D}{2} = 2\pi \times 32.5mm \approx 204.2 \text{ mm/s}$$

5.0 SIMULATION AND RESULTS

5.5.1 Software Simulation

Verification was performed via Proteus to confirm the transition between the LM358 and L293D.

Software Simulation Placeholder
Insert Proteus Circuit Schematic Here

5.5.2 Hardware Implementation

The physical assembly utilized a breadboard architecture on an acrylic chassis with a 12mm sensor gap.



5.5.3 Experimental Results

Round	Line Width	Result	Observation
1	20 mm	Pass	Smooth path; stable.
2	15 mm	Pass	Optimal performance.
3	10 mm	Pass	Successful; high sensitivity.

6.0 CONCLUSION AND RECOMMENDATIONS

The project successfully demonstrated that analog differential steering is a highly responsive method for autonomous tracking. It is recommended to add light shielding around the sensors for improved performance in brightly lit environments.

7.0 BILL OF MATERIALS

- LM358 Dual Op-Amp (1)
- L293D Motor Driver (1)
- IR Sensor Pairs (2)
- 10k Ω Potentiometers (2)
- 60 RPM DC Motors (2)
- 9V Power Source (1)

8.0 REFERENCES

1. Dept. of Electrical and Electronics Engineering. (2026). *EEG 331 Course Manual*. UNILAG.
2. Boylestad, R. L. (2012). *Electronic Devices and Circuit Theory*. Pearson.